RCA Convergence

Introduction

Almost 4 years ago, I began investing and trading stocks. I lost a lot of money early on, but over time I learned how to assess a business's performance, expanded my risk tolerance, and made a profit from a portfolio containing only 5-10 stocks over the last few years.



In my second year of investing I was introduced to the world of algorithmic trading; a world in which Investing Strategists and coders collaborate to develop profitable algorithms that are then used to live trade stocks automatically on the open market. I had one plausible strategy in mind and though it seemed like an efficient way to invest, I was deterred from algorithmic trading, by the fact that I had to write code (mainly Python). At the time, I could barely read code, let alone write up a whole strategy. So I stuck with simple, manual trading.



Over the past few months, I decided to take the plunge into developing the strategy I thought of 2 years ago and after a half-year of strengthening my Python skills, coding the strategy, and extensively testing the strategy on historical data, I now have a functional, profitable algorithm that gives the S&P 500 a run for its money.

How It Works

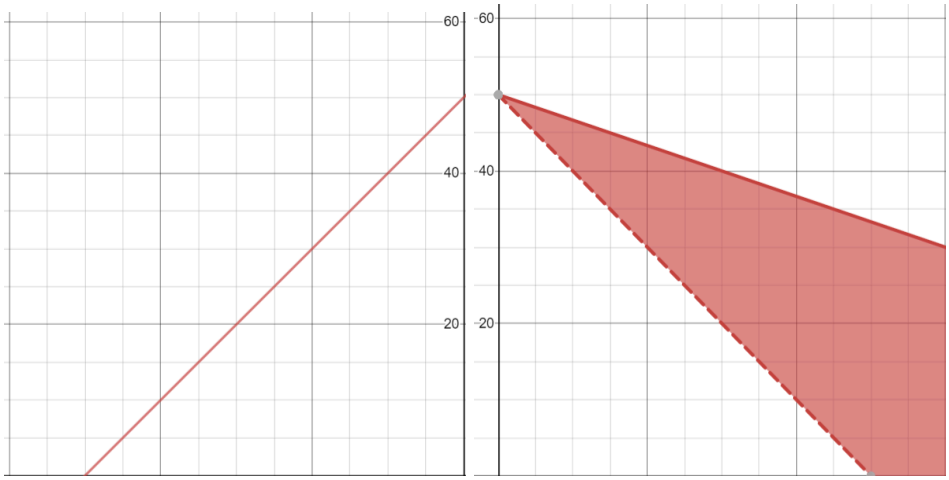
The cornerstone of the strategy is a metric I've coined 'Reflection Covariance'. This metric is calculated by looking at a stock's minute-by-minute close prices of the previous day and reflecting these prices over the y-axes of an (x,y)-plane. So for example, if your minute-prices

are trending upwards for the previous day, the reflection of those prices would trend downward. The set of reflected minute-prices is then compared to the current day's set of minute-prices, by taking the covariance of the pair, i.e. cov(Reflect_Set1, Set2).

$$\mathrm{Cov}\,(X, Y) = \frac{\sum(X_i - \bar{X})(Y_j - \bar{Y})}{n}$$

Formula for Covariance

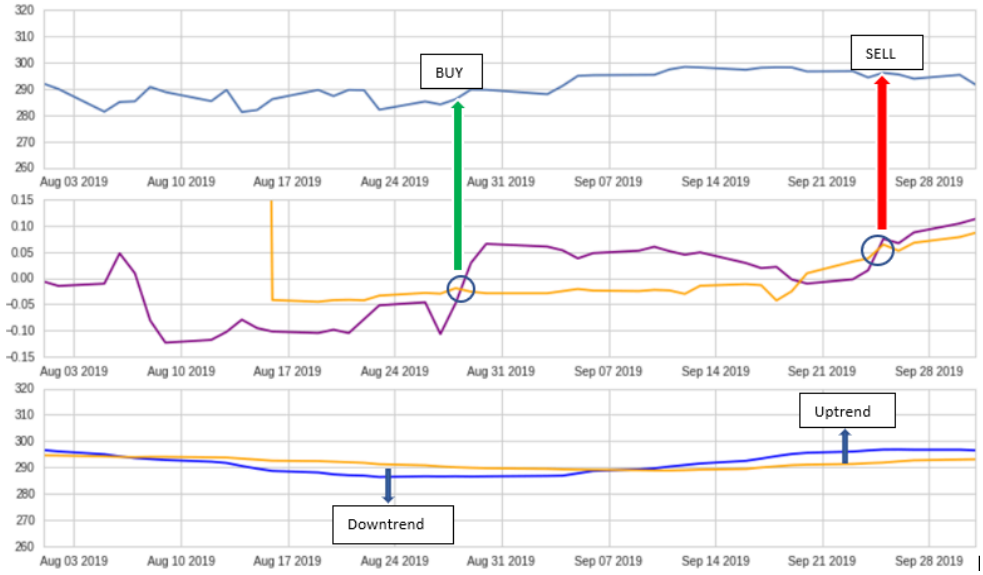
Covariance a measure of the relationship between two random variables (in this case two groups of data points). The metric evaluates how much, or to what extent, the variables change together.



Graph A **Graph B**

On Graph A lies the price movement of a stock after one day of trading. On the Graph B lies the reflection of the previous day's price movement (the dotted line) and the price movement on the current day (the solid line). The covariance between the solid and dotted lines is evaluated and then stored for BUY/SELL signal creation. Each line is made up of 390 points (there are 390 minutes in a single trading day).

The algorithm collects these Reflection Covariances and averages them over time. These RCA's (Reflection Covariance Averages), in addition to closing price SMA's (simple moving averages), generate BUY and SELL signals. This is similar to more well-known strategies (Moving Average Convergence Divergence), in that whenever the short-term RCA crosses over the long-term RCA, the algorithm deems the stock in question a BUY or SELL, depending on whether the stock is in a downtrend (BUY) or uptrend (SELL) based on its SMA's.



Signal Research

In the first graph you'll find price movement. Both circles in the second graph indicate where a signal was detected. A signal is characterized by the short-term RCA crossing over the long-term RCA (the orange line). The first signal is a BUY because of the downtrend, and second signal is a SELL because of the uptrend. Trends are characterized by the short- and long- Moving Averages of price, found in the third graph. If short-term is under long-term (the orange line) you're in a downtrend, and if short-term crosses over long-term you're in an uptrend.

Results

Years since Jan 2008	ALGO 1.3 Annual Returns (%)	SPY Annual Returns (%)
1	-19.6	-38.49
2	37.03	23.45
3	15.56	12.78
4	-2.17	0
5	10.44	13.41
6	30.7	29.6
7	17.27	11.39
8	-2.81	-0.73
9	14.62	9.54
10	15.6	19.42
11	-8.27	-6.24
12	16.92	28.88
Average YOY Return (%)	10.44083333	8.584166667
Std of YOY Return (%)	16.1343442	18.67874118

	Annual Gain/Loss ($)
1	-1043.13
2	1499.63
3	843.02
4	-132.01
5	635.67
6	2067.2
7	1602.47
8	-221.37
9	1550.73
10	1851.53
11	-1177.61
12	2102.74
Avg YOY Gain ($)	798.2391667
Std YOY Gain ($)	1181.463082

From January 1st 2008 to January 1st 2020, the RCA Convergence strategy yields 184.77%, whereas the SPY (a fund that tracks the S&P 500) yields 178.67%. The average annual return of the strategy in that same time span was 10.44% versus the SPY's annual return of 8.58%

(according to https://www.macrotrends.net/2526/sp-500-historical-annual-returns). The strategy was even less risky than trading SPY, with a standard deviation of 16.13 (the standard deviation of SPY annual returns was 18.68). This translates to a gain of $798.24 per year (assuming $5000 in initial capital was deposited at the beginning of January 2008).

Benefits

Aside from the monetary prospects, investing in this strategy takes stock-picking and stock-watching out of the equation. This means consistent gains, without the interference of human error or emotion. The wide array of stocks considered by this algorithm ensures that your investment is well-diversified against risk--more so than the common benchmarks. The algorithm retrieves the top 2000 stocks based on Market Cap and from those takes the top 500 by Dollar Volume. The combination of filters enforces that only the most reputable, liquid stocks are considered.

Call for Action

I intend to invest $5000 to trade according to this strategy, $2000 of which will come from a fund made up of close family members. The algorithm will continue running on Quantopian.com (where it was built) and trading will be conducted on the Interactive Brokers platform. I'm seeking 3-4 more investors that would like to contribute. Trading is set to commence on January 20th.

